Data Analysis with Python

Week 5: Joins, Data Viz, and Econometrics

Justinas Grigaitis, 2026-10-15

Plan for today

Quiz, data analysis, and guest talk (LLMs)

- Feedback (~5 min)
- Quiz (25 min)
- Blitz presentation (~15 min)
  - Joining multiple datasets
  - Visualizing data
  - Basic Econometrics
- Break (~15 min)
- Guest presentation + Q&A (~45 min)

Outline

Data Analysis with Python

- Week 1-2: Python fundamentals
- Week 3: Intro to data analysis, cleaning, and Pandas
- Week 4: Data wrangling and analysis (reshaping and agg/transform)
- **Week 5: quiz, data merging and viz, guest talk on AI engineering**
  - Python fundamentals quiz (20% of grade)
- Week 6: group project idea pitches and sample project walkthrough

Outline

Rest of the semester

- Week 6: student project idea pitches
  - 90 seconds per group - *what*, *why*, and *how*
  - Look for the data to assess project feasibility
  - Try practicing pandas on prelim project data as homework
- Final presentations: Friday, Dec 4, 12:00-16:00
  - 15 groups x (8 min talk + 5 min Q&A) = 195 minutes
  - 2 breaks x 15 min, food provided (TBC) - snacks or pizza

2026 Nobel Prize in Economics

Robert E. Hall and Michael Woodford

“for microeconomic foundations of macroeconomic stabilisation”



Joins, visualization, econometrics

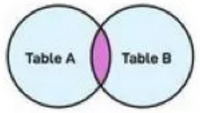
Combining data

5 takeaways from each topic

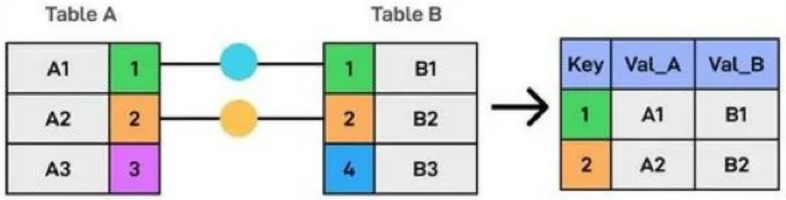
- Use dictionaries when importing multiple datasets
- Use pd.concat() to stack datasets
- Use pd.merge() to join datasets on matching key(s):
  - Left join is what you need 90% of time (9% outer join, <1% cross join)
  - Use validation functions and check shape to make sure the joins work as intended (one-to-one, one-to-many, many-to-one, many-to-many)
  - Be careful about datasets exploding, especially with cross and outer joins

The 4 Types Of SQL Joins

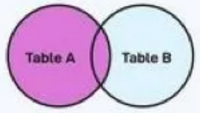
INNER JOIN



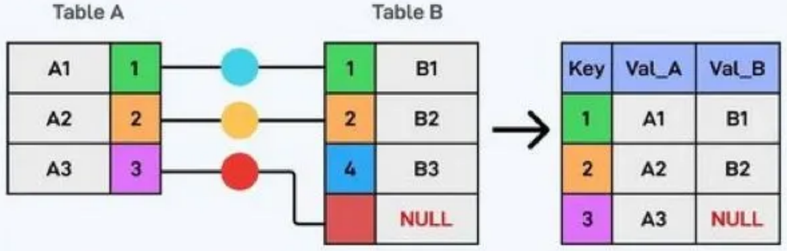

```
SELECT < list_of_columns> FROM
TABLE_A A INNER JOIN TABLE_B B
ON A.KEY = B.KEY
```



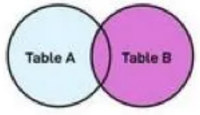
LEFT JOIN



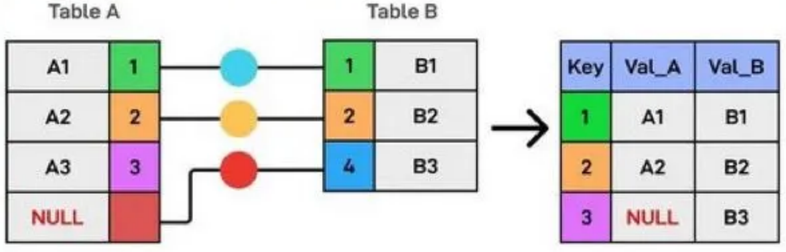

```
SELECT < list_of_columns> FROM
TABLE_A A LEFT JOIN TABLE_B B ON
A.KEY = B.KEY
```



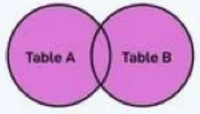
RIGHT JOIN



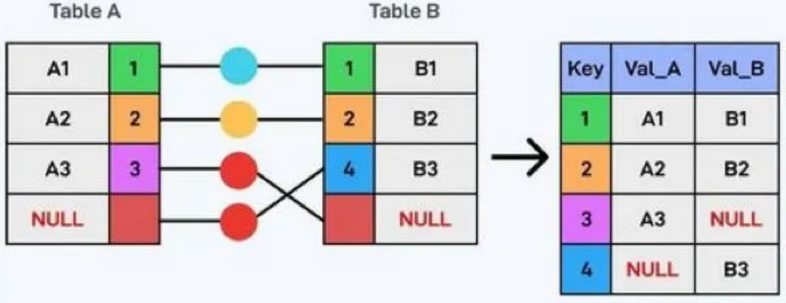

```
SELECT < list_of_columns> FROM
TABLE_A A RIGHT JOIN TABLE_B B
ON A.KEY = B.KEY
```



FULL OUTER JOIN




```
SELECT < list_of_columns> FROM
TABLE_A A FULL OUTER JOIN
TABLE_B B ON A.KEY = B.KEY
```



Data visualization

5 takeaways from each topic

- Get inspired and step up your game
- Data viz is art and science of communication (often very overlooked)
- Start with the main takeaway and design backwards (minimize clutter)
- Adjust the style to the audience (be original within reasonable bounds)
- Best place to use AI (least error-prone, most tedious, hard to memorize)

Econometrics

5 takeaways from each topic

- Python can replicate all of Stata/R (all is math), but might feel more difficult
  - Stata might have more built-in functions (only true for niche Econometrics)
  - Need Python fundamentals (lists, dataframes, numpy arrays, etc.)
- Python is an all-in-one tool (data wrangling -> analysis -> visualization)
- Python is by far the main tool for data science (ML), LLMs, web-scraping, etc.
- Python can be much more computationally effective

Break

Guest speaker, yay!

Aleksander Ficek, Senior Research Scientist @ NVIDIA

https://www.linkedin.com/in/aleksanderficek/

Homework

Homework (~5 hours)

Practice everything as a prep for the final project

- Prepare for the pitch (~1 hour)
  - Brainstorm ideas, find the dataset, practice out loud to fit in 1.5 minutes
- Practice pandas for each topic so far on prelim project data (~3 hours)
  - Cleaning, reshaping, and aggregating (Week 4)
  - Combining data, visualising, running a regression (Week 5)
- Try Streamlit or collaborating on Git (~1 hour)